

Homework 3

Incentive-Centered Design

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Acknowledgements

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2. LLMs referred: ChatGPT and Gemini for doubt clarification

Solutions

1. Myerson's lemma states that if players' utilities are of the form

$$u_i(v_i, \alpha, p_i) = v_i \alpha - p_i$$

where $v_i \in \mathbb{R}_{\geq 0}$ is the private valuation of player i , α is the allocation of player i , and p_i is the payment imposed on player i , then a given allocation rule $\mathbf{x} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is implementable in dominant strategies if and only if $\mathbf{x}$ is monotone. The goal of this problem is to generalize the necessity part of this lemma.

- (a) (10 Points) Suppose that the players' utilities are of the form

$$u_i(v_i, \alpha, p_i) = f(v_i, \alpha) - p_i$$

and that f satisfies the following properties:

- (i) $f(v_i, \cdot)$ is nondecreasing for every v_i , and
- (ii) for every $0 \leq y < z$ and every $0 \leq \alpha < \beta$,

$$f(z, \alpha) - f(y, \alpha) < f(z, \beta) - f(y, \beta).$$

Show that the necessity part of Myerson's lemma continues to hold. I.e., show that if an allocation rule is implementable in dominant strategies, then it is necessarily monotone.

Hint: Follow the same sequence of arguments as in the proof of Myerson's lemma.

- (b) (5 Points) Verify that $f(v_i, \alpha) = v_i \alpha$ (which is usually used) satisfies the two aforementioned properties.
- (c) (5 Points) Construct a function f that is not of the form $f(v, \alpha) = kv\alpha$, $k \in \mathbb{R}$, but still satisfies the two aforementioned properties.

Remark: By imposing another mild condition on f , I believe the sufficiency part can also be generalized. Might be fun to try out when you're bored.

Solution:

- (a) Fix player i and $\mathbf{b}_{-i}$. Let $x_i(v_i)$ denote the allocation. Assume the mechanism is DSIC. Then for any $y < z$:

$$f(z, x_i(z)) - p_i(z) \geq f(z, x_i(y)) - p_i(y)$$

$$f(y, x_i(y)) - p_i(y) \geq f(y, x_i(z)) - p_i(z)$$

Adding both inequalities:

$$f(z, x_i(z)) + f(y, x_i(y)) \geq f(z, x_i(y)) + f(y, x_i(z))$$

Rearranging:

$$f(z, x_i(z)) - f(y, x_i(z)) \geq f(z, x_i(y)) - f(y, x_i(y))$$

Suppose, for contradiction, that $x_i(z) < x_i(y)$. Let

$$\alpha = x_i(z), \quad \beta = x_i(y),$$

so that $\alpha < \beta$.

Now applying property (ii), since $y < z$ and $\alpha < \beta$, we obtain

$$f(z, \alpha) - f(y, \alpha) < f(z, \beta) - f(y, \beta).$$

Substituting $\alpha = x_i(z)$ and $\beta = x_i(y)$,

$$f(z, x_i(z)) - f(y, x_i(z)) < f(z, x_i(y)) - f(y, x_i(y)),$$

which contradicts

$$f(z, x_i(z)) - f(y, x_i(z)) \geq f(z, x_i(y)) - f(y, x_i(y)).$$

Hence,

$$x_i(z) \geq x_i(y),$$

so the allocation rule is monotone.

(b) Let $f(v, \alpha) = v\alpha$.

(i) For fixed $v \geq 0$, $f(v, \alpha) = v\alpha$ is increasing in α .

(ii) For $y < z$ and $\alpha < \beta$:

$$f(z, \beta) - f(y, \beta) = (z - y)\beta, \quad f(z, \alpha) - f(y, \alpha) = (z - y)\alpha.$$

Since $(z - y) > 0$ and $\beta > \alpha$:

$$(z - y)\alpha < (z - y)\beta.$$

(c) Consider:

$$f(v, \alpha) = v\alpha + \alpha^2.$$

(i) For fixed v ,

$$\frac{\partial f}{\partial \alpha} = v + 2\alpha \geq 0,$$

so $f(v, \alpha)$ is increasing in α .

(ii) For $y < z$:

$$f(z, \alpha) - f(y, \alpha) = (z - y)\alpha,$$

which is increasing in α . Hence for $\alpha < \beta$,

$$(z - y)\alpha < (z - y)\beta.$$

Thus the function satisfies both properties and is not of the form $kv\alpha$.

2. (10 Points) When proving Myerson's lemma in class, we assumed that the allocation function is differentiable. This problem deals with an extension wherein the allocation function is not differentiable.

Consider a single-item auction wherein the seller wishes to allocate the item to the player who values it the most. Hence, the allocation function is

$$x_i(b_i, \mathbf{b}_{-i}) = \begin{cases} 1 & \text{if } b_i > \max\{\mathbf{b}_{-i}\}, \\ 0 & \text{Otherwise.} \end{cases} \quad (1)$$

Using Myerson's lemma, derive the payment rule $\mathbf{p}$ that would make $(\mathbf{x}, \mathbf{p})$ DSIC. Assume boundary condition $p_i(b_i = 0, \mathbf{b}_{-i}) = 0$ for all i and all $\mathbf{b}_{-i}$.

Hint: Start from equation (8) of the proof of Myerson's lemma (Lecture 20, Slide 10).

Solution:

Fix bidder i and $\mathbf{b}_{-i}$, and let $m = \max\{\mathbf{b}_{-i}\}$.

The allocation rule is:

$$x_i(b_i, \mathbf{b}_{-i}) = \begin{cases} 1 & \text{if } b_i > m, \\ 0 & \text{otherwise.} \end{cases}$$

From Myerson's lemma, for $z > y$:

$$y[x_i(z) - x_i(y)] \leq p_i(z) - p_i(y) \leq z[x_i(z) - x_i(y)].$$

Case 1: $z \leq m$

Then $x_i(z) = x_i(y) = 0$, so:

$$p_i(z) = p_i(y).$$

Using $p_i(0) = 0$, we get $p_i(b_i) = 0$ for all $b_i \leq m$.

Case 2: $y < m < z$

Then $x_i(z) = 1$, $x_i(y) = 0$, so:

$$y \leq p_i(z) \leq z.$$

Taking $y \rightarrow m^-$:

$$m \leq p_i(z) \leq z.$$

For DSIC, the payment for all winning bids $z > m$ must be identical; otherwise a bidder could profitably deviate by lowering their bid while still winning. Hence:

$$p_i(z) = m.$$

Conclusion:

$$p_i(b_i, \mathbf{b}_{-i}) = \begin{cases} \max\{\mathbf{b}_{-i}\} & b_i > \max\{\mathbf{b}_{-i}\}, \\ 0 & \text{otherwise.} \end{cases}$$

3. (3+3+4=10 Points) Compute the revenue-maximizing price when there is only one potential buyer, a single item, and the buyer's valuation density f is

(i) $f(v) = \begin{cases} 1 & 0 \leq v \leq 1, \\ 0 & \text{else.} \end{cases}$

(ii) $f(v) = \begin{cases} 4v & 0 \leq v \leq \frac{1}{2}, \\ 4 - 4v & \frac{1}{2} \leq v \leq 1. \end{cases}$

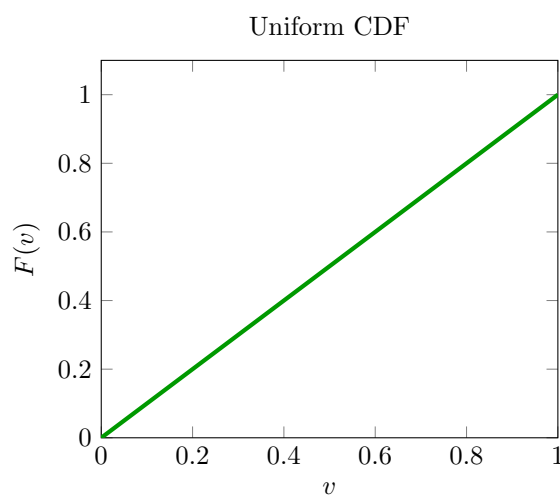
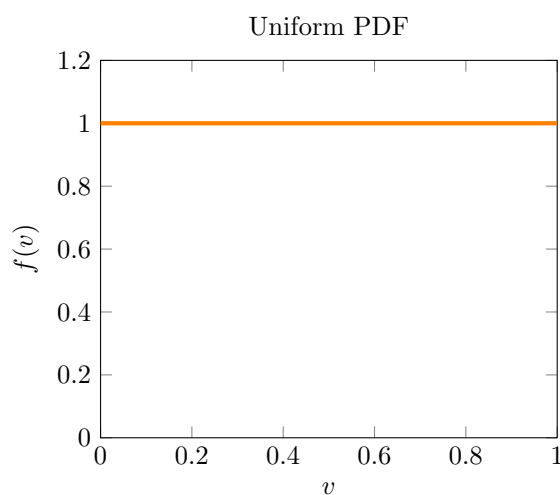
(iii) $f(v) = \begin{cases} \frac{\pi}{2} \sin(\pi v) & 0 \leq v \leq 1 \\ 0 & \text{else.} \end{cases}$

In each case, before even you begin solving, use qualitative arguments to guess the range in which the optimal price should lie (greater than, less than, or equal to $\frac{1}{2}$). Then solve to find the exact price.

Solution: Fair price is obtained by maximizing $R(p) = p(1 - F(p))$.

(i)

$$F(v) = \begin{cases} 0 & v < 0 \\ v & 0 \leq v \leq 1 \\ 1 & v > 1 \end{cases}$$



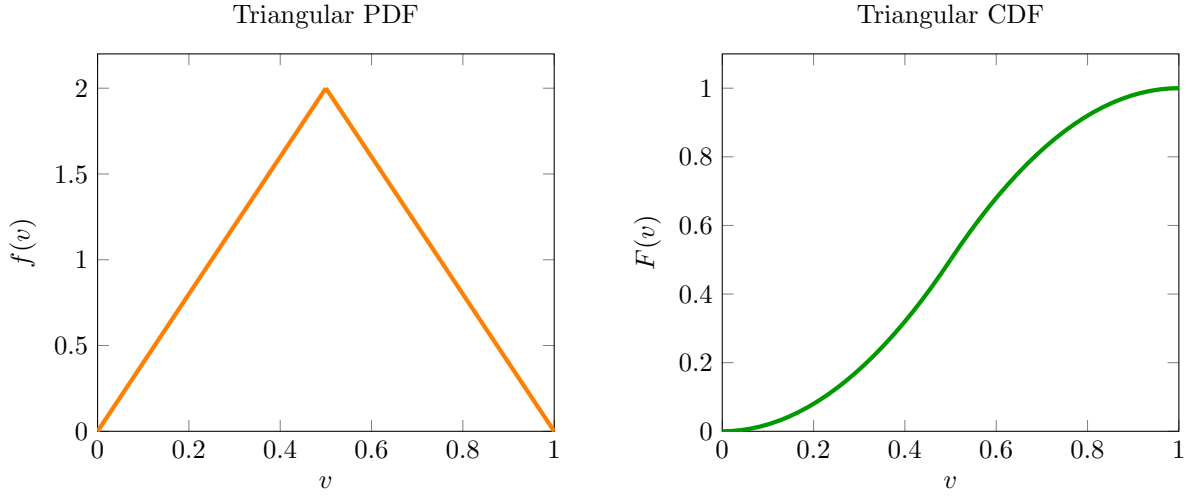
By symmetry, we expect $p^* = \frac{1}{2}$.

$$R(p) = p(1 - p), \quad R'(p) = 1 - 2p = 0 \Rightarrow p = \frac{1}{2}$$

$p^* = \frac{1}{2}$

(ii)

$$F(v) = \begin{cases} 0 & v < 0 \\ 2v^2 & 0 \leq v \leq \frac{1}{2} \\ 4v - 2v^2 - 1 & \frac{1}{2} \leq v \leq 1 \\ 1 & v > 1 \end{cases}$$



Since mass is concentrated near the center, we expect $p^* < \frac{1}{2}$.

For $p \leq \frac{1}{2}$:

$$R(p) = p(1 - 2p^2) = p - 2p^3$$

$$R'(p) = 1 - 6p^2 = 0 \Rightarrow p = \frac{1}{\sqrt{6}}$$

For $p \geq \frac{1}{2}$:

$$R(p) = p(2 - 4p + 2p^2) \Rightarrow R'(p) = 2 - 8p + 6p^2$$

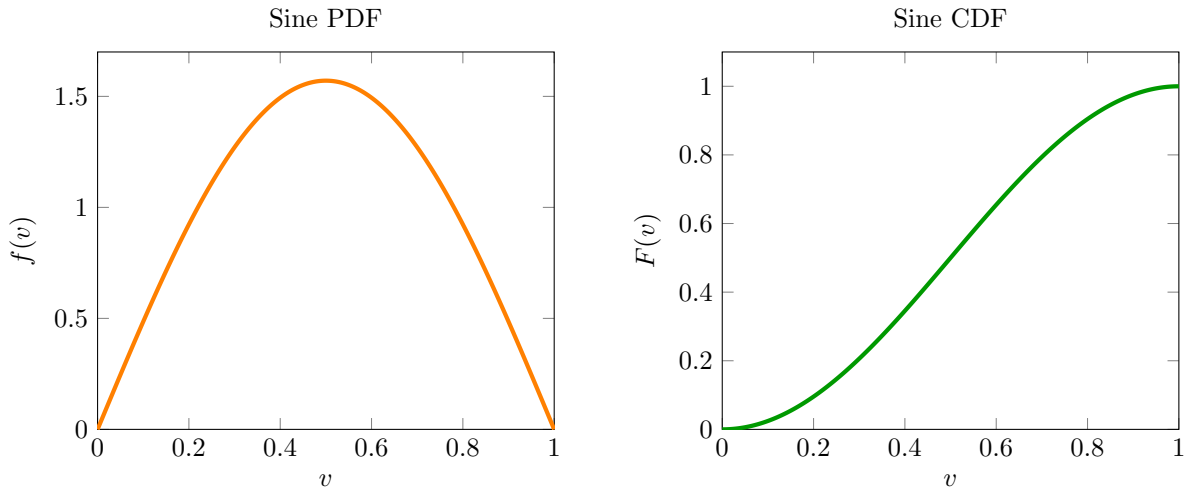
The critical points are $p = \frac{1}{3}$ and $p = 1$. Both lie outside the region $p \geq \frac{1}{2}$, hence are not valid here.

Thus, the maximizer is:

$$p^* = \frac{1}{\sqrt{6}} \approx 0.408$$

(iii)

$$F(v) = \begin{cases} 0 & v < 0 \\ \frac{1 - \cos(\pi v)}{2} & 0 \leq v \leq 1 \\ 1 & v > 1 \end{cases}$$



Again, mass is concentrated around the center, suggesting $p^* < \frac{1}{2}$.

$$R(p) = \frac{p(1 + \cos(\pi p))}{2}$$

$$R'(p) = \frac{1 + \cos(\pi p)}{2} - \frac{\pi p \sin(\pi p)}{2} = 0$$

This equation has a unique solution in $[0, 1]$, which can be verified numerically. hence this critical point is the unique maximizer.

Solving numerically:

$$p^* \approx 0.37$$

4. (5 Points) Which of the following probability distributions are regular? Justify your answer.

(i) $Unif([a, b])$.

(ii) $Exp(\lambda)$.

Solution:

A distribution is said to be regular if its virtual value function

$$\phi(v) = v - \frac{1 - F(v)}{f(v)}$$

is increasing in v .

(i) **Uniform distribution** $Unif([a, b])$

$$f(v) = \frac{1}{b-a}, \quad F(v) = \frac{v-a}{b-a}$$

$$1 - F(v) = \frac{b-v}{b-a}$$

$$\frac{1 - F(v)}{f(v)} = b - v$$

$$\phi(v) = v - (b - v) = 2v - b$$

$$\phi'(v) = 2 > 0$$

Hence, the uniform distribution is **regular** as the corresponding virtual value function is monotonically increasing.

(ii) **Exponential distribution** $Exp(\lambda)$

$$f(v) = \lambda e^{-\lambda v}, \quad F(v) = 1 - e^{-\lambda v}$$

$$1 - F(v) = e^{-\lambda v}$$

$$\frac{1 - F(v)}{f(v)} = \frac{e^{-\lambda v}}{\lambda e^{-\lambda v}} = \frac{1}{\lambda}$$

$$\phi(v) = v - \frac{1}{\lambda}$$

$$\phi'(v) = 1 > 0$$

Hence, the exponential distribution is **regular** as the corresponding virtual value function is monotonically increasing.

5. (10 Points) A seller wishing to maximize revenue has a single indivisible item and two prospective buyers. Buyer 1 has valuation density $Unif([0, 10])$ for the item and Buyer 2 has valuation density $Exp(2)$. Compute the auction that the seller should employ. I.e., for each possible bid profile of the players, determine who (if any) wins the item and at what price. Justify your answer using Myerson's lemma.

Solution:

We apply Myerson's lemma. In a single-parameter setting, the revenue-maximizing auction allocates the item to the bidder with the highest *non-negative virtual value*, and charges the corresponding threshold payment.

Step 1: Virtual valuations

For Buyer 1 ($Unif([0, 10])$):

$$f_1(v) = \frac{1}{10}, \quad F_1(v) = \frac{v}{10}$$

$$\phi_1(v_1) = v_1 - \frac{1 - F_1(v_1)}{f_1(v_1)} = v_1 - (10 - v_1) = 2v_1 - 10$$

For Buyer 2 ($Exp(2)$):

$$f_2(v) = 2e^{-2v}, \quad F_2(v) = 1 - e^{-2v}$$

$$\phi_2(v_2) = v_2 - \frac{1 - F_2(v_2)}{f_2(v_2)} = v_2 - \frac{1}{2}$$

Step 2: Interpretation

$$\phi_1(v_1) \geq 0 \iff v_1 \geq 5, \quad \phi_2(v_2) \geq 0 \iff v_2 \geq \frac{1}{2}$$

Step 3: Allocation rule

Allocate to bidder with highest non-negative virtual value.

Step 4: Payment rule

Payments are obtained by solving the threshold condition defining the minimum bid required to win.

- Buyer 1 threshold solves:

$$2v_1' - 10 = \max \left\{ v_2 - \frac{1}{2}, 0 \right\}$$

$$p_1 = \begin{cases} \frac{v_2}{2} + \frac{19}{4} & v_2 \geq \frac{1}{2} \\ 5 & v_2 < \frac{1}{2} \end{cases}$$

- Buyer 2 threshold solves:

$$v_2' - \frac{1}{2} = \max \{ 2v_1 - 10, 0 \}$$

$$p_2 = \begin{cases} 2v_1 - \frac{19}{2} & v_1 \geq 5 \\ \frac{1}{2} & v_1 < 5 \end{cases}$$

- No allocation: $p_1 = p_2 = 0$

The item is allocated to bidder 1 if

$$2v_1 - 10 \geq \max \left\{ v_2 - \frac{1}{2}, 0 \right\},$$

and to bidder 2 if

$$v_2 - \frac{1}{2} \geq \max \{ 2v_1 - 10, 0 \}.$$

Otherwise, the item is not allocated.

6. (5 Points) Consider a single-item auction with n buyers. The buyers' valuations are independent and identically distributed random variables with F being the cumulative distribution function of buyer 1's valuation. Assume that F is regular. Prove that of all DSIC auctions that allocate the item, the Vickrey auction maximizes revenue.

Remark: Recall that we appealed to this result in proving a result about prior-free auctions.

Solution:

Let $(\mathbf{x}, \mathbf{p})$ be any DSIC mechanism that always allocates the item. By Myerson's lemma, its expected revenue equals the expected virtual surplus:

$$\mathbb{E} \left[\sum_{i=1}^n \phi(v_i) x_i(\mathbf{v}) \right],$$

where $\phi(v) = v - \frac{1-F(v)}{f(v)}$.

Since the mechanism always allocates the item, we have $\sum_{i=1}^n x_i(\mathbf{v}) = 1$ for all $\mathbf{v}$, so the revenue becomes:

$$\mathbb{E} [\phi(v_{w(\mathbf{v})})],$$

where $w(\mathbf{v})$ is the winning bidder.

Since F is regular, $\phi(v)$ is increasing in v . Hence for every valuation profile $\mathbf{v}$,

$$\max_i \phi(v_i) = \phi \left(\max_i v_i \right).$$

Therefore, for any allocation rule x ,

$$\sum_{i=1}^n \phi(v_i) x_i(\mathbf{v}) \leq \max_i \phi(v_i),$$

with equality if and only if the item is allocated to a bidder with the highest valuation.

Taking expectations,

$$\mathbb{E} \left[\sum_i \phi(v_i) x_i(\mathbf{v}) \right] \leq \mathbb{E} [\phi(\max_i v_i)].$$

Thus, the revenue is maximized by allocating the item to the bidder with the highest valuation.

Among DSIC mechanisms with this allocation rule, Myerson's lemma implies that the payment rule is uniquely determined (up to constants) and equals the threshold bid required to win. For the highest bidder, this threshold is the second-highest bid.

Hence, the Vickrey (second-price) auction maximizes revenue among all DSIC mechanisms that always allocate the item.

7. (5 Points) Consider a multi-parameter environment wherein the valuations of all players are nonnegative for all alternatives. In the lecture, we showed that the VCG mechanism is DSIC, IR, and efficient. Prove that the VCG mechanism is weakly budget-balanced.

Solution:

Let $\omega^*(\mathbf{b}) \in \arg \max_{\omega} \sum_j b_j(\omega)$ be the welfare-maximizing outcome, and for each i let

$$\omega_{-i}^* \in \arg \max_{\omega} \sum_{j \neq i} b_j(\omega)$$

be the welfare-maximizing outcome when player i is removed.

The VCG payment of player i is

$$p_i(\mathbf{b}) = \sum_{j \neq i} b_j(\omega_{-i}^*) - \sum_{j \neq i} b_j(\omega^*(\mathbf{b})).$$

Summing over all i ,

$$\begin{aligned} \sum_i p_i(\mathbf{b}) &= \sum_i \sum_{j \neq i} b_j(\omega_{-i}^*) - \sum_i \sum_{j \neq i} b_j(\omega^*(\mathbf{b})) \\ &= \sum_i \sum_{j \neq i} b_j(\omega_{-i}^*) - (n-1) \sum_j b_j(\omega^*(\mathbf{b})). \end{aligned}$$

For each i , by optimality of ω_{-i}^* for players $-i$,

$$\sum_{j \neq i} b_j(\omega_{-i}^*) \geq \sum_{j \neq i} b_j(\omega^*(\mathbf{b})).$$

Summing this inequality over all i gives

$$\sum_i \sum_{j \neq i} b_j(\omega_{-i}^*) \geq \sum_i \sum_{j \neq i} b_j(\omega^*(\mathbf{b})) = (n-1) \sum_j b_j(\omega^*(\mathbf{b})).$$

Hence,

$$\sum_i p_i(\mathbf{b}) \geq 0.$$

Therefore, the VCG mechanism is weakly budget-balanced.

8. (5+10+5+5=25 Points) The VCG mechanism that we saw in class assumed that the valuations of all players are nonnegative for every outcome. The first part of this problem shows that the incentive compatibility of the VCG mechanism continues to hold even when valuations are negative. The later parts of the problem applies the result of the first part to establish a fundamental impossibility result pertaining to bilateral trade, viz., efficient bilateral trade cannot occur without external subsidy — a result that lies at the heart of tensions in international trade.

- (i) Let Ω be a finite set of outcomes and denote by $v_i(\omega) \in \mathbb{R}$ the valuation of player i for outcome $\omega \in \Omega$. The valuation can be any real number; not just nonnegative. Denote by $\mathbf{b}_i$ the bid of player i , $i \in \{1, \dots, n\}$. Show that if the social planner employs the decision rule

$$\omega^*(\mathbf{b}_1, \dots, \mathbf{b}_n) = \arg \max_{\omega \in \Omega} \sum_{i=1}^n b_i(\omega) \quad (2)$$

and payment rule

$$p_i(\mathbf{b}_i, \mathbf{b}_{-i}) = f_i(\mathbf{b}_{-i}) - \sum_{j \neq i} b_j(\omega^*(\mathbf{b}_1, \dots, \mathbf{b}_n)) \quad (3)$$

for all $i \in \{1, \dots, n\}$, where f_i is any arbitrary function, then truthful bidding by all players is a dominant strategy. Also show that at equilibrium, the outcome chosen by the social planner maximizes social welfare.

Remark: Conversely, every DSIC mechanism has a payment rule that is of the form (3) for some f_i s.

- (ii) Two parties are interested in trading an indivisible good. The seller values the good at v_s and the buyer values the good at v_b . The valuations are private to the players. There are two possible outcomes: (i) trade occurs (Y), or (ii) trade does not occur (N). The valuations of the players for all outcomes are as specified in the following table.

	Y	N
<i>Seller</i>	$-v_s$	0
<i>Buyer</i>	v_b	0

A third person, say a market maker, wishes to implement the following outcome function:

- (Efficiency) If $v_b \geq v_s$, outcome Y must be chosen, and if $v_b < v_s$, outcome N must be chosen. I.e., the good must find itself in the hands of the person who values it the most.

Design a DSIC mechanism which implements the aforementioned outcome function. The mechanism must also satisfy the following two properties:

- (a) (Individual Rationality) Both parties must obtain a nonnegative utility regardless of what the outcome is,
 - (b) In the event that no trade occurs, both parties pay 0.
- (iii) Does the mechanism designed in Part (ii) satisfy budget balance?
- (iv) Let's relax the requirement that both parties pay 0 in the event that no trade occurs. Can we now design a mechanism that satisfies budget balance? Justify.

Solution:

(i)

Fix player i and $\mathbf{b}_{-i}$. The mechanism chooses:

$$\omega^*(\mathbf{b}_i, \mathbf{b}_{-i}) = \arg \max_{\omega \in \Omega} \sum_{j=1}^n b_j(\omega).$$

The utility of player i is:

$$u_i(v_i, \mathbf{b}_i, \mathbf{b}_{-i}) = v_i(\omega^*) - p_i(\mathbf{b}_i, \mathbf{b}_{-i}).$$

Substituting the payment rule,

$$p_i(\mathbf{b}_i, \mathbf{b}_{-i}) = f_i(\mathbf{b}_{-i}) - \sum_{j \neq i} b_j(\omega^*),$$

we obtain:

$$\begin{aligned} u_i &= v_i(\omega^*) - \left[f_i(\mathbf{b}_{-i}) - \sum_{j \neq i} b_j(\omega^*) \right] \\ &= v_i(\omega^*) + \sum_{j \neq i} b_j(\omega^*) - f_i(\mathbf{b}_{-i}). \end{aligned}$$

Observe that $f_i(\mathbf{b}_{-i})$ does not depend on b_i . Hence, player i can influence utility only through the choice of outcome ω^* .

Thus, player i seeks to choose b_i so as to maximize:

$$v_i(\omega^*) + \sum_{j \neq i} b_j(\omega^*).$$

Since ω^* is chosen to maximize $\sum_j b_j(\omega)$, player i achieves this by reporting truthfully, i.e.,

$$b_i = v_i.$$

Hence, truthful bidding is a dominant strategy.

Efficiency:

At truthful bidding, $b_i = v_i$ for all i , so:

$$\omega^* = \arg \max_{\omega \in \Omega} \sum_{i=1}^n v_i(\omega),$$

which maximizes total social welfare.

(ii) Design of DSIC, IR mechanism

There are two outcomes: trade (Y) and no trade (N). Valuations are:

$$v_b(Y) = v_b, \quad v_b(N) = 0, \quad v_s(Y) = -v_s, \quad v_s(N) = 0.$$

The efficient outcome is:

$$\omega^*(v_b, v_s) = \begin{cases} Y & \text{if } v_b \geq v_s, \\ N & \text{otherwise.} \end{cases}$$

Using the generalized VCG mechanism:

$$p_i(\mathbf{b}) = f_i(\mathbf{b}_{-i}) - \sum_{j \neq i} b_j(\omega^*).$$

Choice of f_i :

- For the buyer:

$$f_b(b_s) = \max_{\omega} b_s(\omega) = \max\{-v_s, 0\} = 0.$$

- For the seller:

$$f_s(b_b) = \max_{\omega} b_b(\omega) = \max\{b_b, 0\} = b_b.$$

Payments:

- If trade occurs ($v_b \geq v_s$):

$$p_b = 0 - (-v_s) = v_s, \quad p_s = b_b - b_b = 0.$$

- If no trade:

$$p_b = 0, \quad p_s = 0.$$

Utilities:

- If trade:

$$u_b = v_b - v_s \geq 0, \quad u_s = -v_s + v_s = 0.$$

- If no trade:

$$u_b = 0, \quad u_s = 0.$$

Thus, the mechanism is DSIC (by part (i)), efficient, individually rational, and satisfies the requirement that payments are zero when no trade occurs.

(iii) Budget balance

Total payments are:

$$p_b + p_s = \begin{cases} v_s & \text{if trade occurs,} \\ 0 & \text{otherwise.} \end{cases}$$

Thus,

$$p_b + p_s \geq 0,$$

so the mechanism is weakly budget-balanced. However, it is not exactly budget-balanced, since the mechanism collects a surplus of v_s when trade occurs.

(iv) Possibility of budget balance

Suppose there exists a mechanism that is simultaneously:

- Efficient,
- DSIC,
- Individually rational,
- Budget balanced.

Let $t(v_b, v_s)$ denote the transfer from the buyer to the seller whenever trade occurs.

By efficiency, trade must occur iff

$$v_b \geq v_s.$$

Since the mechanism is budget balanced, the buyer's payment equals the seller's receipt.

Now consider the utilities when trade occurs.

For the buyer:

$$u_b = v_b - t(v_b, v_s).$$

Individual rationality implies

$$v_b - t(v_b, v_s) \geq 0,$$

hence

$$t(v_b, v_s) \leq v_b.$$

For the seller:

$$u_s = t(v_b, v_s) - v_s.$$

Individual rationality implies

$$t(v_b, v_s) - v_s \geq 0,$$

hence

$$t(v_b, v_s) \geq v_s.$$

Therefore, whenever trade occurs,

$$v_s \leq t(v_b, v_s) \leq v_b.$$

Now fix the seller's valuation v_s . Since the mechanism is DSIC, the buyer must face a threshold price independent of his own report; otherwise he could manipulate the payment while still obtaining trade.

Hence there exists some threshold price $p(v_s)$ such that trade occurs iff

$$v_b \geq p(v_s).$$

Similarly, fixing the buyer's valuation v_b , DSIC for the seller implies there exists a threshold $q(v_b)$ such that trade occurs iff

$$v_s \leq q(v_b).$$

Efficiency requires trade iff

$$v_b \geq v_s.$$

Thus we must have

$$p(v_s) = v_s \quad \text{and} \quad q(v_b) = v_b.$$

But then the buyer effectively faces payment v_s and the seller effectively receives v_b . Since these depend on the other agent's private report, each agent can profitably manipulate the transfer while preserving the trade decision, contradicting DSIC.

Hence no mechanism can simultaneously satisfy efficiency, DSIC, individual rationality, and budget balance.

This is precisely the impossibility captured by the Myerson–Satterthwaite theorem.